1. EOQ

1.1. Base

Parameters

D	Demand rate	items / time
h	Holding cost (rate)	% / item / time
H	Holding cost	$ / item / time
c	Cost	$ / item
S	Setup cost	$ / order
L	Lead time	time

Assumptions

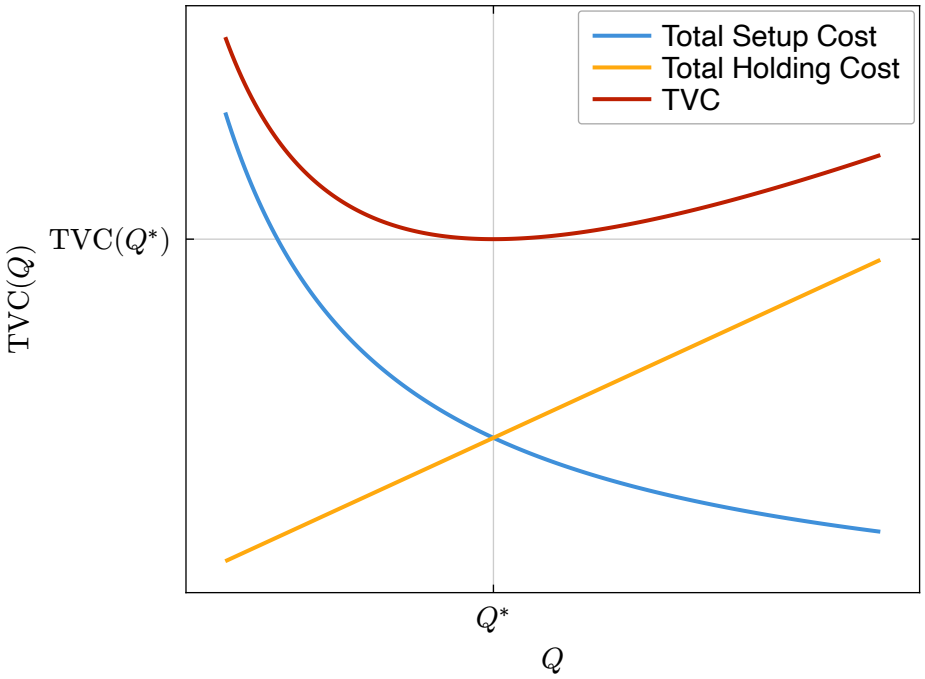
- Demand D is **known** and **constant**
- Lead time L is **known** and **constant**
- Replenishment is **instantaneous**
- Setup costs S is **constant**
- Unit cost c is **constant**
- Holding cost H is a **constant**
- No stockouts or backorders allowed

Start with:

$$\mathrm{TC}(Q) = cD + S\frac{D}{Q} + H\frac{Q}{2}$$

The cD term is not a function of Q so:

$$\mathrm{TVC}(Q) = S\frac{D}{Q} + H\frac{Q}{2}$$



Take the derivative of $\mathrm{TVC}(Q)$:

$$\frac{\mathrm{d}\ \mathrm{TVC}(Q)}{\mathrm{d}Q} = -\frac{SD}{Q^2} + \frac{H}{2}$$

Confirm it is a minimum rather than a maximum:

$$\frac{\mathrm{d}^2\ \mathrm{TVC}}{\mathrm{d}Q^2} = \frac{2SD}{Q^3} > 0 \text{ for } Q > 0$$

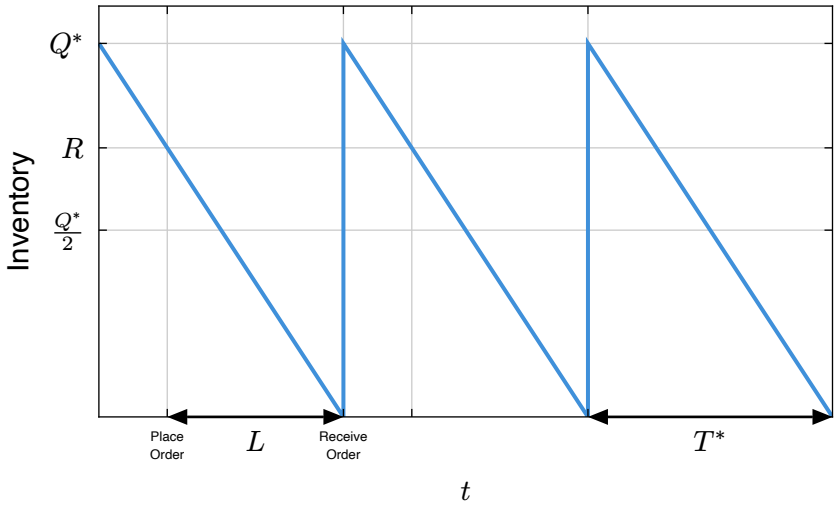
Set it to 0 to get the minimum (optimal order quantity / batch size):

$$-\frac{SD}{Q^2} + \frac{H}{2} = 0$$

$$Q^* = \sqrt{\frac{2DS}{H}}$$

Backfill Q^* into the $\mathrm{TVC}(Q)$ to get the optimal Total Variable cost corresponding to the Q^*:

$$\begin{aligned}\mathrm{TVC}(Q^*) &= S\frac{D}{Q^*} + H\frac{Q^*}{2} \\ &= \sqrt{2DSH} \\ \mathrm{TC}(Q^*) &= cD + \mathrm{TVC}(Q^*) \\ &= cD + \sqrt{2DSH}\end{aligned}$$



1.2. Optimal Cycle Time t^*

$$\begin{aligned}t^* &= \frac{Q^*}{D} \\ &= \frac{\sqrt{\frac{2DS}{H}}}{D} \\ &= \frac{1}{D}\sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2S}{DH}}\end{aligned}$$

$$t^* = \frac{1}{N^*}$$

1.3. Optimal Order Frequency N^*

$$\begin{aligned} N^* &= \frac{D}{Q^*} \\ &= \frac{D}{\sqrt{\frac{2DS}{H}}} \\ &= \sqrt{\frac{DH}{2S}} \end{aligned}$$

$$N^* = \frac{1}{t^*}$$

1.4. Cycle Inventory

Cycle inventory is the average amount of stock you're carrying purely because you order in batches instead of receiving supply continuously, one unit at a time, exactly as it's consumed.

$$\frac{Q^*}{2}$$

1.5. Sensitivity

$$x = \frac{Q}{Q^*}$$

Recall that:

$$-\frac{SD}{Q^{*2}} + \frac{H}{2} = 0 \implies \frac{SD}{Q^{*2}} = \frac{H}{2}$$

Multiply both sides by Q^*:

$$\frac{SD}{Q^*} = \frac{HQ^*}{2}$$

Since the two terms are equal to each other:

$$\mathrm{TVC}(Q^*) = 2 \cdot S\frac{D}{Q^*}, \quad \text{so } S\frac{D}{Q^*} = \frac{\mathrm{TCV}(Q^*)}{2}$$

Similarly:

$$\mathrm{TVC}(Q^*) = 2 \cdot H\frac{Q^*}{2}, \quad \text{so } H\frac{Q^*}{2} = \frac{\mathrm{TCV}(Q^*)}{2}$$

Therefore:

$$S\frac{D}{Q^*} = H\frac{Q^*}{2} = \frac{\mathrm{TVC}(Q^*)}{2}$$

Substitute $Q = xQ^*$ into $\mathrm{TVC}(Q) = \frac{SD}{Q} + \frac{HQ}{2}$:

$$\begin{aligned}\mathrm{TVC}(xQ^*) &= \frac{SD}{xQ^*} + \frac{HxQ^*}{2} \\ &= \frac{1}{x} \cdot \frac{SD}{Q^*} + x \cdot \frac{HQ^*}{2}\end{aligned}$$

Simplifying, using the identity above:

$$\begin{aligned}\mathrm{TVC}(xQ^*) &= \frac{1}{x} \cdot \frac{\mathrm{TVC}(Q^*)}{2} + x \cdot \frac{\mathrm{TVC}(Q^*)}{2} \\ &= \frac{\mathrm{TVC}(Q^*)}{2}\left(x + \frac{1}{x}\right)\end{aligned}$$

Dividing both sides by $\mathrm{TVC}(Q^*)$:

$$\frac{\mathrm{TVC}(Q)}{\mathrm{TVC}(Q^*)} = \frac{1}{2}\left(x + \frac{1}{x}\right)$$

1.6. Lead Time L

$$R = dL$$

1.7. Power of 2 2^k

Recall the sensitivity formula:

$$\frac{1}{2}\left(x + \frac{1}{x}\right) \quad \text{where } x = \frac{Q}{Q^*}$$

Plugging in $x = \frac{Q}{Q^*} = \sqrt{2}$:

$$\frac{1}{2}\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = \frac{1}{2} \cdot \frac{3\sqrt{2}}{2} = \frac{3\sqrt{2}}{4} \approx 1.0607$$

Cycle time obtained by picking the best integer k

$$\frac{\mathrm{TVC}\left(2^{k^*}\right)}{\mathrm{TVC}(T^*)} \leq 1.06$$

1.8. EPQ

Parameters

D	Demand rate	items / time
P	Production rate	items / time
h	Holding cost (rate)	% / item / time
H	Holding cost	$ / item / time
c	Cost	$ / item
S	Setup cost	$ / order
L	Lead time	time
T_p	Product time per cycle	

T_d	Down time per cycle	
$I_{\max}$	Maximum inventory	items

$$P \gg D$$

Assumptions

- Demand D is **known** and **constant**
- Lead time L is **known** and **constant**
- Replenishment is **instantaneous**
- Setup costs S is **constant**
- Unit cost c is **constant**
- Holding cost H is a **constant**
- No stockouts or backorders allowed

$$T = \frac{Q}{D}$$

$$T_p = \frac{Q}{P}$$

$$T_d = T - T_p = \frac{Q}{D} - \frac{Q}{P}$$

$$I_{\max} = T_p(P - D) = Q\left(1 - \frac{D}{P}\right)$$

$$T_d = \frac{I_{\max}}{D} = \frac{Q}{D}\left(1 - \frac{D}{P}\right)$$

$$1 - \frac{D}{P} = T_d\frac{D}{Q} \quad (DH_{i(\text{cycles / year})}$$

$$T = T_p + T_d = \frac{Q}{D}$$

$$\bar{I}_{\max} = \frac{I_{\max}}{2}$$

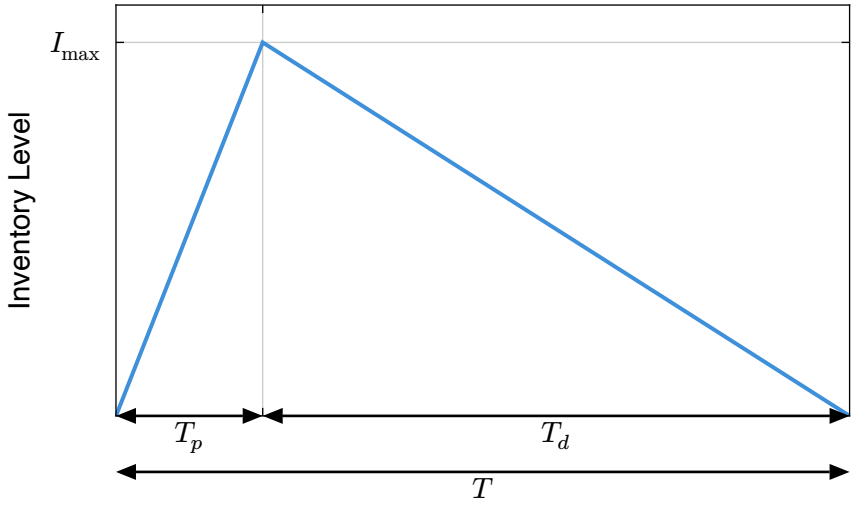
$$\text{TVC}(Q) = S\frac{D}{Q} + H\frac{Q}{2}\left(1 - \frac{D}{P}\right)$$

$$\text{EPQ} = Q^* = \sqrt{\frac{2DS}{H(1 - \frac{D}{P})}}$$

$$\text{TVC}(Q) = \sqrt{2DSH\left(1 - \frac{D}{P}\right)}$$

If $P \gg D$ then $\left(1 - \frac{D}{P}\right) \to 1$ meaning we have EOQ.

We scale the holding cost H



1.9. Quantity Discounts (All Unit Discount)

Assumptions

- Demand D is **known** and **constant**
- Lead time L is **known** and **constant**
- Replenishment is **instantaneous**
- Setup costs S is **constant**
- Unit cost c is **constant**
- Holding cost H is a **constant**
- No stockouts or backorders allowed

Breakpoints: $q_0, q_1, ..., q_m$

If order Q is between q_i and q_{i+1}, each unit is obtained at cost c_i

The cost decreases as the quantity ordered increases: $c_0 \geq c_1 \geq ... \geq c_{m-1}$

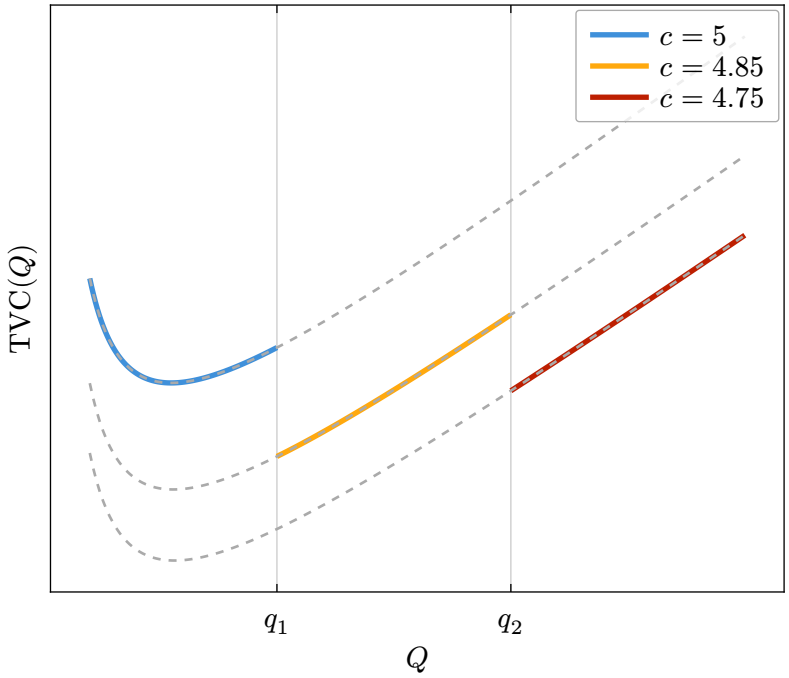
Minimize sum of **purchase**, order, holding costs

TVC now includes purchase cost

$$\mathrm{TVC}_i(Q) = Dc_i + \frac{D}{Q}S + \frac{Q}{2}hc_i, \quad \text{for } q_i \leq Q \leq q_{i+1}$$

$$\frac{\mathrm{d}\ \mathrm{TVC}}{\mathrm{d}Q} = -\frac{SD}{Q^2} + \frac{hc_i}{2}$$

$$Q_i^* = \sqrt{\frac{2DS}{hc_i}}$$



- Step 1: Calculate the EOQ for the lowest price. If it is feasible (i.e.,

this order quantity is within the range for that price), then stop. This is the optimal lot size. Calculate total variable cost (TVC) for this lot size.
- Step 2: If the EOQ is not feasible, calculate the TVC for this price

and the smallest quantity for that price.
- Step 3: Calculate the EOQ for the next lowest price. If it is feasible,

stop and calculate the TVC for that quantity and price.
- Step 4: Compare the TVC for Steps 2 and 3. Choose the quantity

corresponding to the lowest TVC.
- Step 5: If the EOQ in Step 3 is not feasible, repeat Steps 2, 3, and 4

until a feasible EOQ is found.

Start from lowes price, evaluate EOQ for each c_i:
- Case 1: If $q_i \leq Q_i < q_{i+1}$, then Q_i is feasible and

$$\mathrm{TVC}_i = \frac{D}{Q_i} S + \frac{Q_i}{2} h c_i + D c_i$$

- Case 2: If $Q_i < q_i$, then choose q_i and

$$\mathrm{TVC}_i = \frac{D}{q_i} S + \frac{q_i}{2} h c_i + D c_i$$

Select Q^* with the lowest TVC

1.10. Quantity Discounts (Marginal Unit Discount)

1.11. Aggregation

Parameters

D_i	Demand rate	items / time
P	Production rate	items / time
h	Holding cost (rate)	% / item / time
H	Holding cost	$ / item / time
c	Cost	$ / item
S	Setup cost	$ / order
L	Lead time	time
T_p	Product time per cycle	
T_d	Down time per cycle	
$I_{\max}$	Maximum inventory	items